

# Customer Shopping Behavior Analysis

3,900 purchases · 18 columns · exploratory analysis (Python + SQL + Power BI)



# Dataset Summary

## Scope

Rows: 3,900 · Columns: 18

## Key Features

Demographics, Purchase details, Behavior, Shipping

## Missing Data

37 nulls in Review Rating



Clean Data

EDITABLE STROKE

# Python — Preparation & Engineering

01

## Load & Inspect

pandas: df.info(), .describe()

02

## Missing Values

Imputed Review Rating by category median

03

## Standardize & Feature Eng.

snake\_case columns; age\_group; purchase\_frequency\_days

04

## DB Integration

Loaded cleaned DF into PostgreSQL

```
import pandas as pd

# Load CSV file and reading the data
df = pd.read_csv('laptops.csv')
```

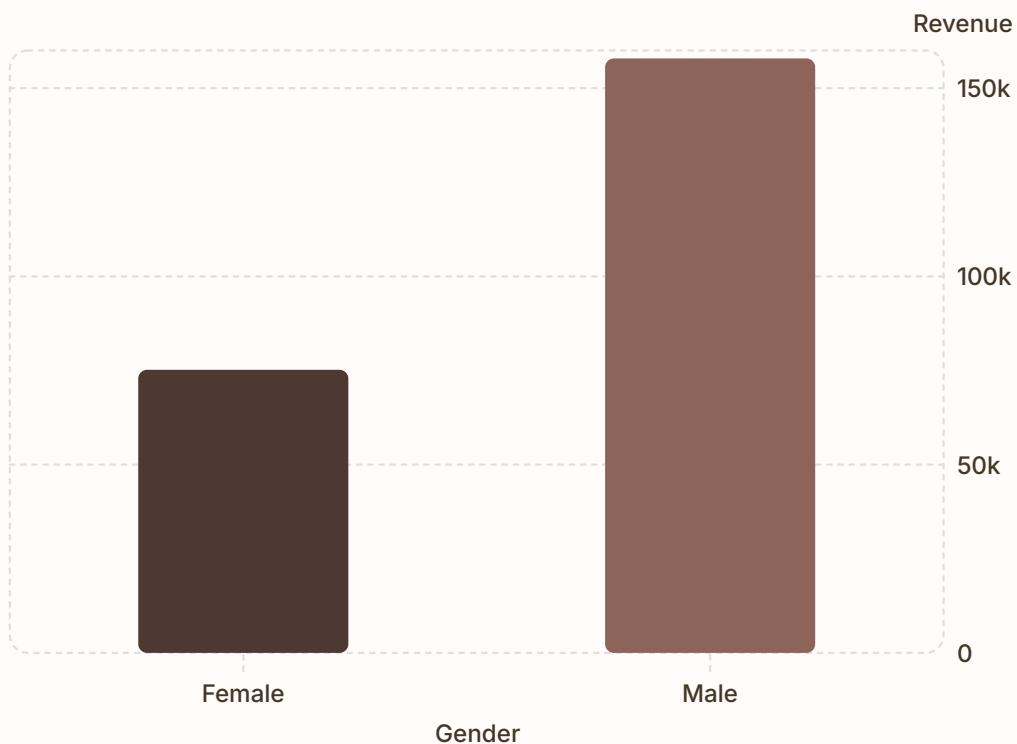
| Model   | Price | In stock |
|---------|-------|----------|
| Macbook | 1400  | Yes      |
| Macbook | 900   | Yes      |
| HP      | NaN   | No       |
| Macbook | 2500  | 6        |
| Macbook | 1800  | No       |

```
52
53
54 USE DatabaseName
55 GO
56 CREATE PROCEDURE ProcedureName
57     @FirstName type,
58     @LastName type
59 AS
60 |
61 //Your SQL query here
62 Select FirstName, LastName
63 FROM Employee
64 WHERE FirstName = @FirstName
65 GO
66 DECLARE @ContactCode INT
67     ContactCode = 0
```

# SQL Analysis — Key Questions

- Revenue by gender
- Discount users with high spend
- Top products by rating
- Shipping type comparison
- Subscribers vs. non-subscribers

# Revenue by Gender



Male customers generated higher total revenue (157,890) vs Female (75,191).

- ❏ Consider targeted outreach to high-value male segments while improving female conversion.

# Top Products & Ratings



**Gloves**

Avg Rating 3.86



**Sandals**

3.84



**Boots**

3.82



**Hat**

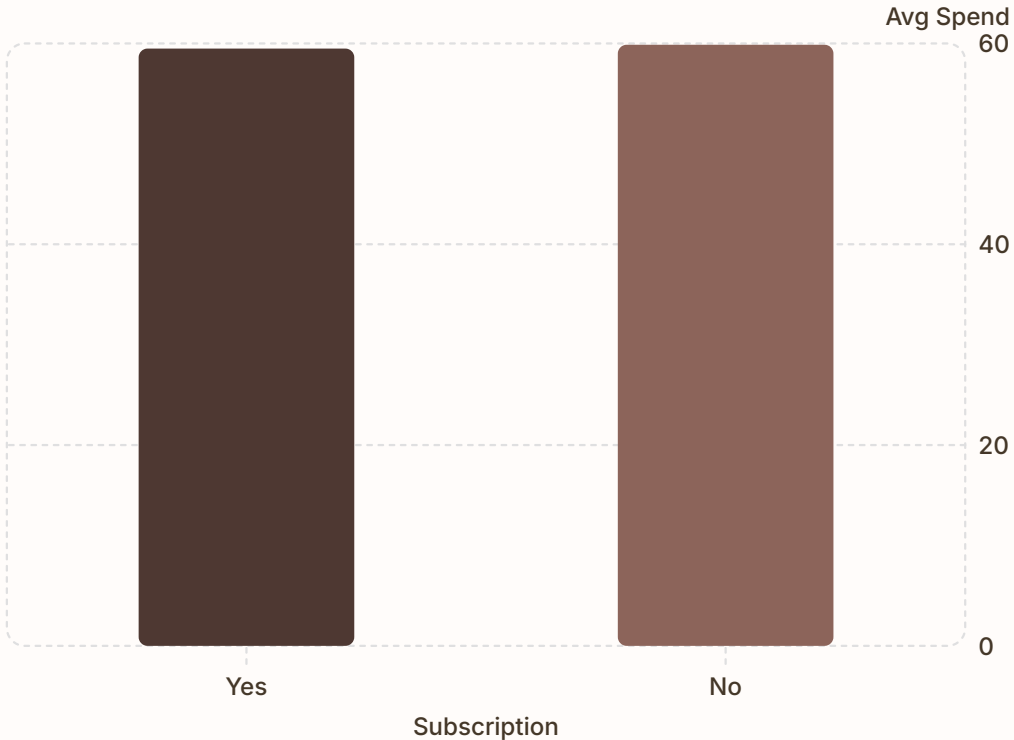
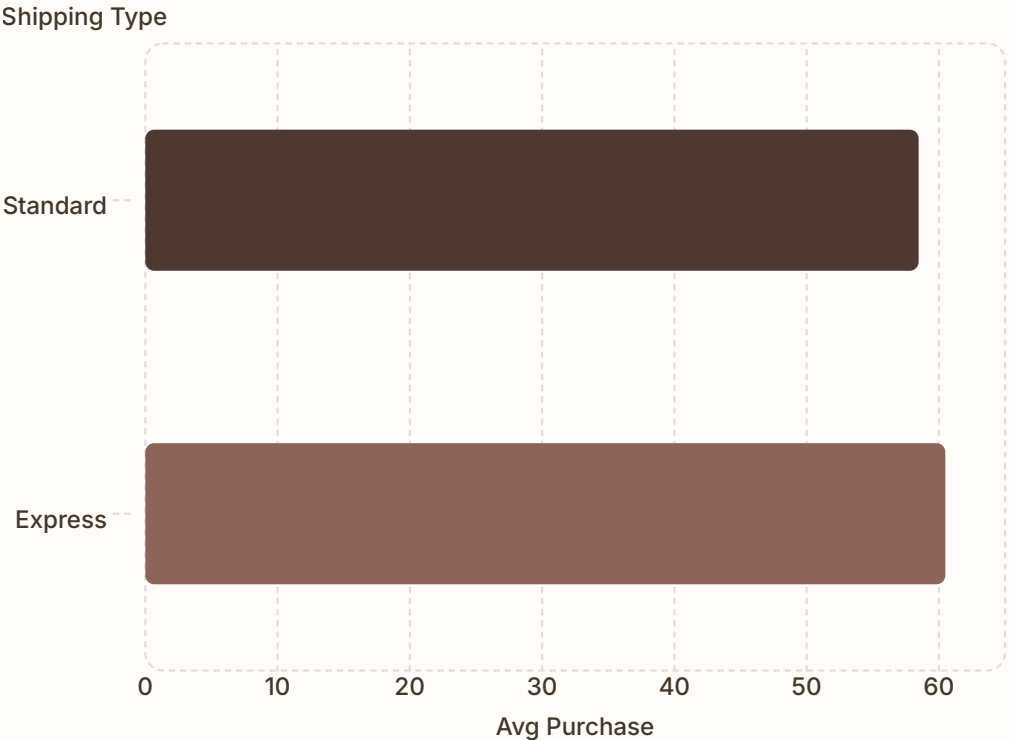
3.80



**Skirt**

3.78

# Shipping & Subscription Insights



Subscribers: 1,053 customers · Total revenue 62,645  
Non-subscribers: 2,847 customers · Total revenue 170,436

# Customer Segments & Repeat Buyers

## 1 Loyal

3,116 customers

## 2 Returning

701 customers

## 3 New

83 customers

## Repeat Buyers vs Subscriptions

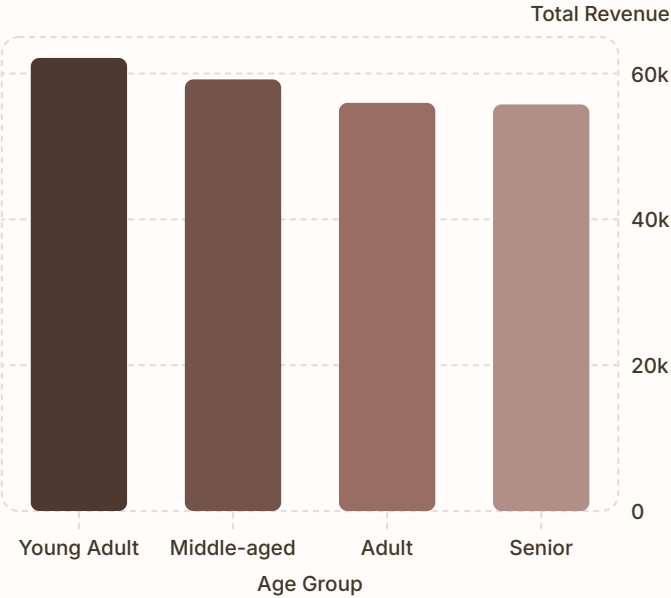
Repeat buyers (>5 purchases): 958 subscribed? check correlation

 Recommendation: convert Returning → Loyal via loyalty program.



# Age Groups & Revenue

High-revenue age groups: Young Adult & Middle-aged.



### Where's Your Target Audience Spending Time?



### Gen Z

- YouTube
- Instagram
- TikTok
- Prefers shorts, temporary and engaging formats like IG Stories (24-hour lifespan)



### Millennials

- Facebook
- YouTube
- Instagram
- X
- Pinterest
- Authentic and relatable content.
- Visual-heavy media like photos and videos.



### Gen X

- Facebook
- YouTube
- Instagram
- X
- Pinterest
- Visual-first content like photos and videos.
- Values informative content



### Baby Boomers

- Facebook (96% use it weekly).
- YouTube
- Instagram
- Visual content like photos and videos.
- Trustworthy and straightforward brand messages.



### Cross-Generational Trends

- All generations agree on the value of visual content (photos and videos) over text-based posts.
- YouTube is a universally popular platform across all age groups.

# Recommendations — Tech Professional

## **Boost Subscriptions**

Promote exclusive subscriber benefits

## **Loyalty Program**

Reward repeat buyers → increase loyalty

## **Discount Strategy**

Review policy to protect margins

## **Product Positioning**

Highlight top-rated & best-selling items

## **Targeted Marketing**

Focus on high-revenue age groups & express-shipping users